### (3) Derivative of the logarithmic function

**The natural logarithm of $x$ is denoted by $\log_e x$ or $\log x$ or $\ln x$**

$$\text{Let } y = f(x) = \log x$$

$$\text{Now } f(x+\Delta x) = \log(x+\Delta x) \text{ and}$$

$$f(x+\Delta x) - f(x) = \log(x+\Delta x) - \log x$$

$$= \log\left(\frac{x+\Delta x}{x}\right).$$

$$= \log\left(1+\frac{\Delta x}{x}\right).$$

$$\frac{f(x+\Delta x)-f(x)}{\Delta x} = \frac{\log\left(1+\frac{\Delta x}{x}\right)}{\Delta x}.$$

We know that $\lim\limits_{\alpha\to 0}\dfrac{\log(1+\alpha k)}{\alpha} = k\lim\limits_{\alpha\to 0}\dfrac{\log(1+k\alpha)}{k\alpha} = k$

Therefore, $\lim\limits_{\Delta x\to 0}\dfrac{f(x+\Delta x)-f(x)}{\Delta x} = \lim\limits_{\Delta x\to 0}\dfrac{\log\left(1+\frac{\Delta x}{x}\right)}{\Delta x} = \dfrac{1}{x}.$

That is, $\boxed{\dfrac{d}{dx}(\log x) = \dfrac{1}{x}}$.

**Corollary 10.3**

$$\text{If } y = f(x) = \log_a x \text{ then } f'(x) = \frac{1}{(\log a)x}.$$

We have $f(x) = \log_a x = \log_a e \times \log_e x = (\log_a e)\log x$

Therefore, $\dfrac{d}{dx}(f(x)) = \dfrac{d}{dx}(\log_a e \times \log x)$

$= (\log_a e)\dfrac{d}{dx}(\log x)$ (by constant multiple rule)

$= \log_a e.\dfrac{1}{x}$ (or) $\dfrac{1}{(\log a)x}$.

### (4) Derivative of the exponential function

$$\text{Let } y = a^x$$

$$\text{Then } f(x+\Delta x) - f(x) = a^{x+\Delta x} - a^x$$

$$= a^x(a^{\Delta x} - 1) \text{ and}$$



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